

ITN276 – Project II

Expert Forensic Report

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Case: Suspected Illegal Firearm Activity & Possible Theft

Tool Used: E3 Forensic Platform (JBL VM)

Date: December 1, 2025

1. Forensic Process & Methodology

1.1 Evidence Acquisition

The first step I took in analyzing the provided suspect image was to **load the disk image into E3 Forensic Platform on the JBL virtual machine**. This ensured that the exhibit was opened in a forensically sound environment, maintaining file system integrity and preventing accidental modification of evidence.

No changes were made to the source image. All analysis was performed using E3's built-in viewers, search functions, and keyword extraction tools.

1.2 Preservation Methods

- The original image remained write-protected throughout the investigation.
- E3 automatically generated hash values for verification.
- All exported artifacts were stored inside the *Project Report* folder generated by E3.

1.3 Tools Used

- **E3 Forensic Platform** – primary tool for browsing file systems, keyword extraction, and metadata review.
- **E3 Chrome Keyword Manager Viewer** – used to analyze browser history artifacts.
- **E3 File Viewer** – used to inspect documents, hidden data, and image attachments.
- **RTF viewer** – for reading embedded notes and communication logs.

1.4 Search Strategy

Because the image contained large volumes of irrelevant personal files, I created a targeted method:

1. **Reviewed Chrome's search terms and browsing history**
2. **Searched file system for:**
 - a. Keywords: *gun, firearm, sell, Nevada, Virginia, transfer, permit, steghide, hex, hxd, Guatemala, Manassas*
 - b. Known file types related to steganography or document editing
3. **Reviewed Recently Used Files** for behavioral context
4. **Examined Downloads Folder** for tools or illicit media
5. **Reviewed User Documents and Messages** for communication related to transactions

This systematic process isolated all relevant forensic artifacts.

2. Evidence Discovery & Explanation

Below is the evidence discovered, its relevance, and what it suggests about user activity.

2.1 Browser Search History

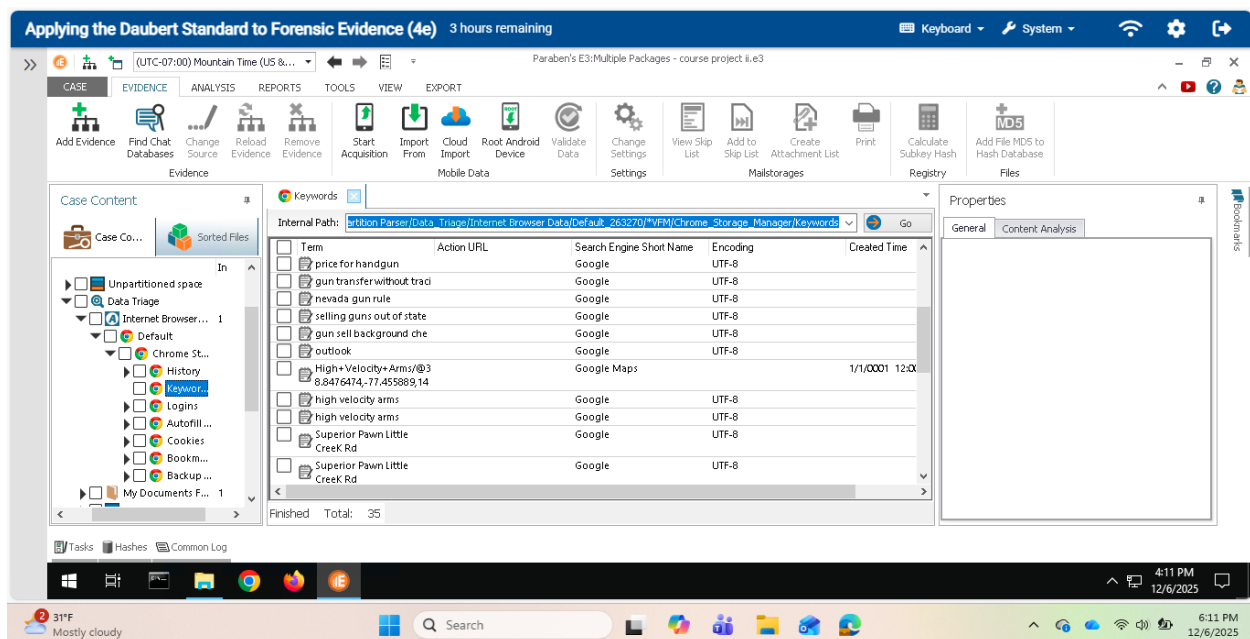
Queries Related to Gun Purchases & Sales

- *"price for handgun"*

- “selling guns out of state”
- “nevada gun rule”
- “gun transfer without traci(n)g”
- “can I sell a firearm to a nonlicense[n]e who is a resident...”
- “background checks and/or purchase permit”

Relevance:

These searches demonstrate **intent to buy, sell, or transfer a firearm**, including questions specifically about **avoiding traceable transfers**, which suggests knowledge of and intent to bypass legal requirements.

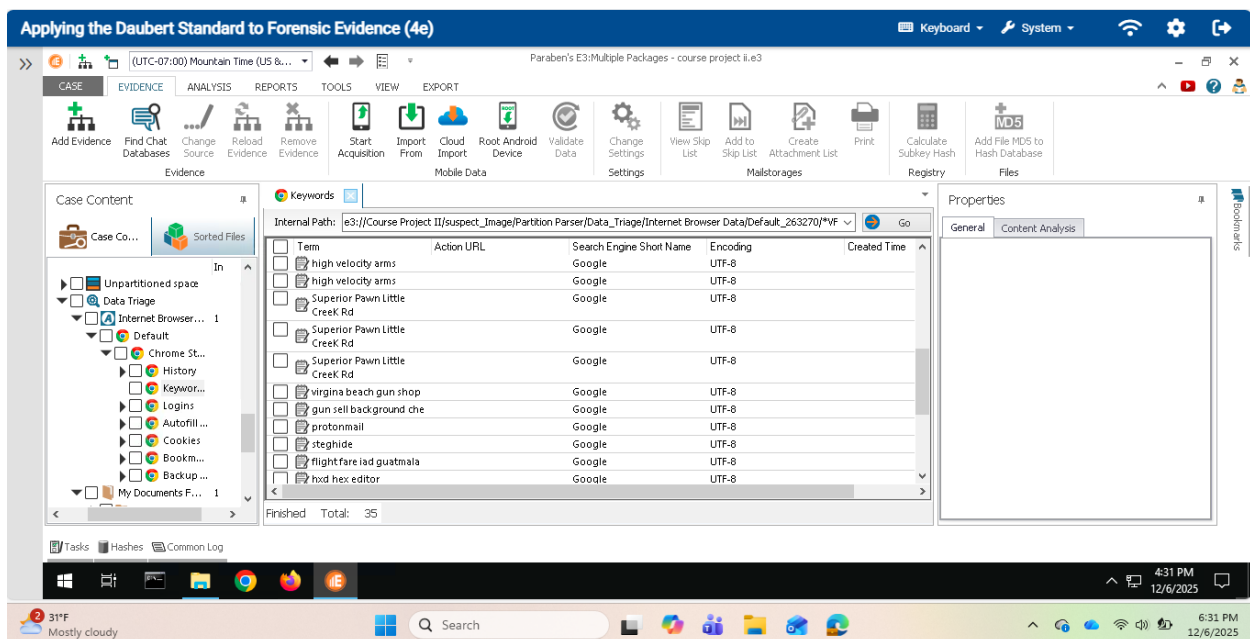
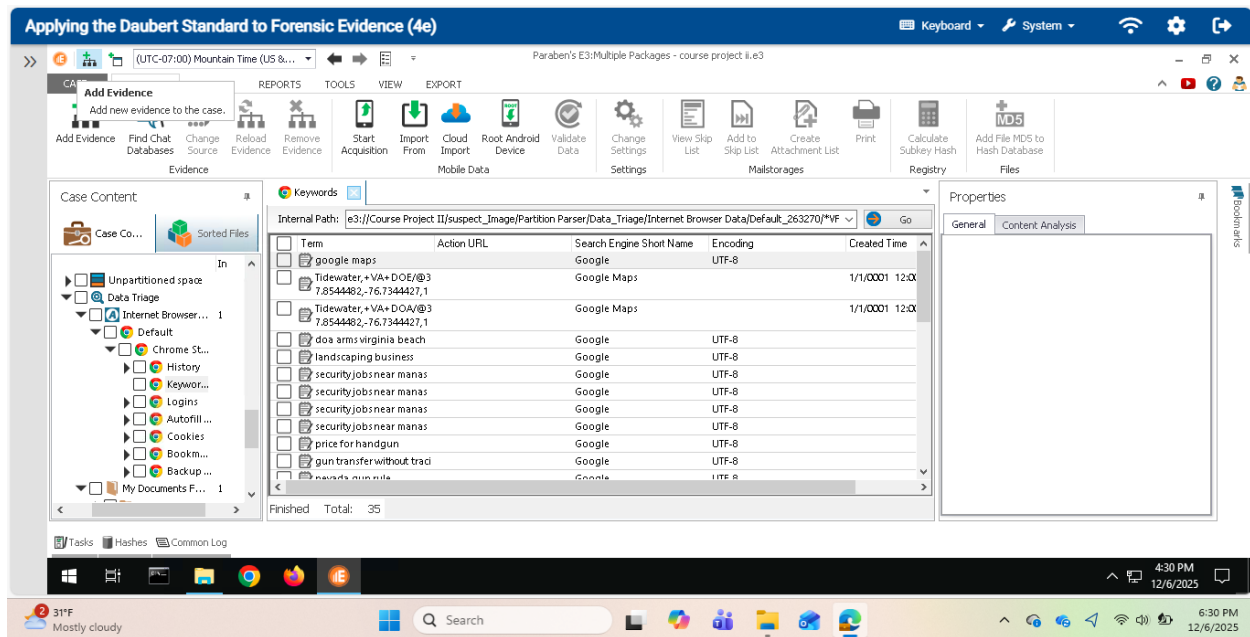


2.2 Searches for Local Gun Shops & Travel Routes

- Google Maps searches for:
 - Manassas, VA
 - DOA Arms
 - High Velocity Arms
 - Superior Pawn – Little Creek Road
- Coordinates appear inside Chrome keyword database.

Relevance:

User appears to be mapping distance between **Manassas and Virginia Beach**, potentially to complete firearm transactions.



2.3 Downloads Folder – Potential Evidence

Steghide Executable

File found:

- **steghide-0.5.1-win32**
- Related searches: “steghide,” “hxd hex editor,” “7zip download”

Relevance:

Steghide is a steganography tool used to hide data inside image/audio files. Combined with the presence of gun images, this strongly suggests attempts to conceal illicit information.

The screenshot displays the Paraben's E3 software interface, titled "Applying the Daubert Standard to Forensic Evidence (4e)". The interface is divided into several sections: a top menu bar, a toolbar with various analysis tools, a left sidebar for "Case Content", a central "Internal Path" view, and a right sidebar for "Properties".

The "Internal Path" view shows a list of files and folders. The file "steghide-0.5.1-win32" is selected, and its properties are displayed in the right sidebar. The properties include:

- File System Attributes: Archive (False), Compressed (False), Directory (True), Encrypted (False), Hidden (False), No Indexed Content (False), Offline (False), Read only (False), Reparse Point (False), Sparse file (False), System (False).
- File System Attributes: steghide-0.5.1-win32: Allocated size (bytes) (0), Complete file size (0), Creation time (1/10/2023 10:55:27 PM).

The bottom status bar shows the system clock as 4:32 PM on 12/6/2025.

The screenshot displays the Paraben's E3 software interface, titled "Applying the Daubert Standard to Forensic Evidence (4e)". The interface is divided into several sections: a top menu bar, a toolbar with various analysis tools, a left sidebar for "Case Content", a central "Internal Path" view, and a right sidebar for "Properties".

The "Internal Path" view shows a list of keywords. The keyword "steghide" is selected, and its properties are displayed in the right sidebar. The properties include:

- Keyword: steghide
- Action URL: Google
- Search Engine Short Name: UTF-8
- Encoding: UTF-8
- Created Time: 1/10/2023 10:55:27 PM

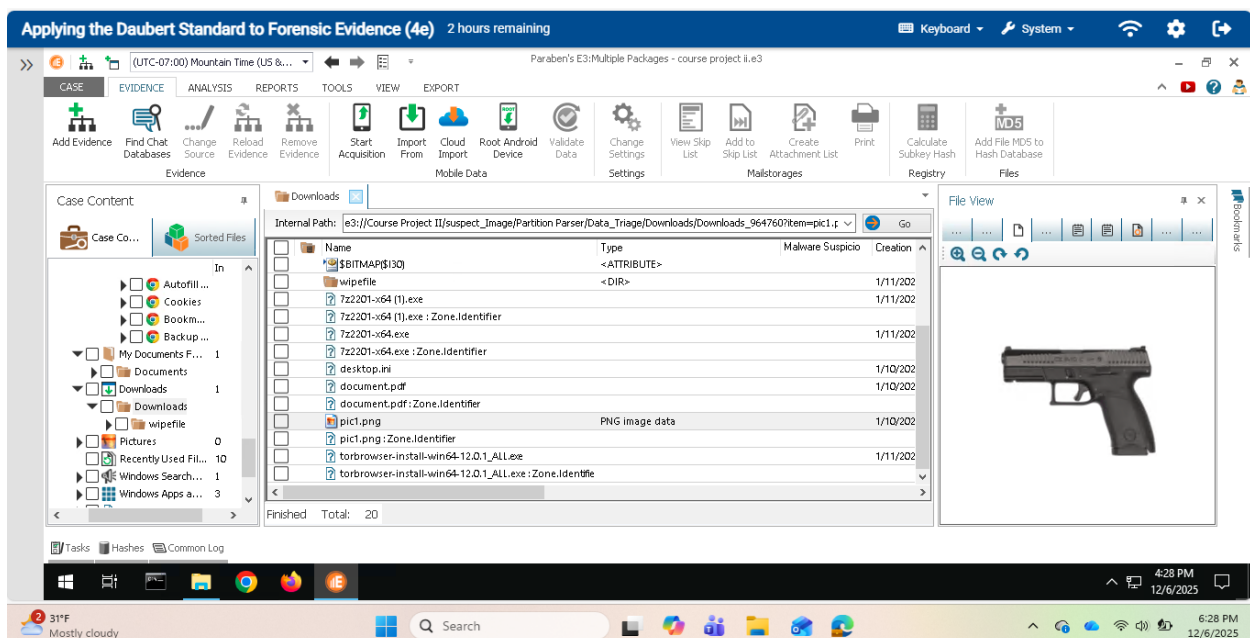
The bottom status bar shows the system clock as 4:33 PM on 12/6/2025.

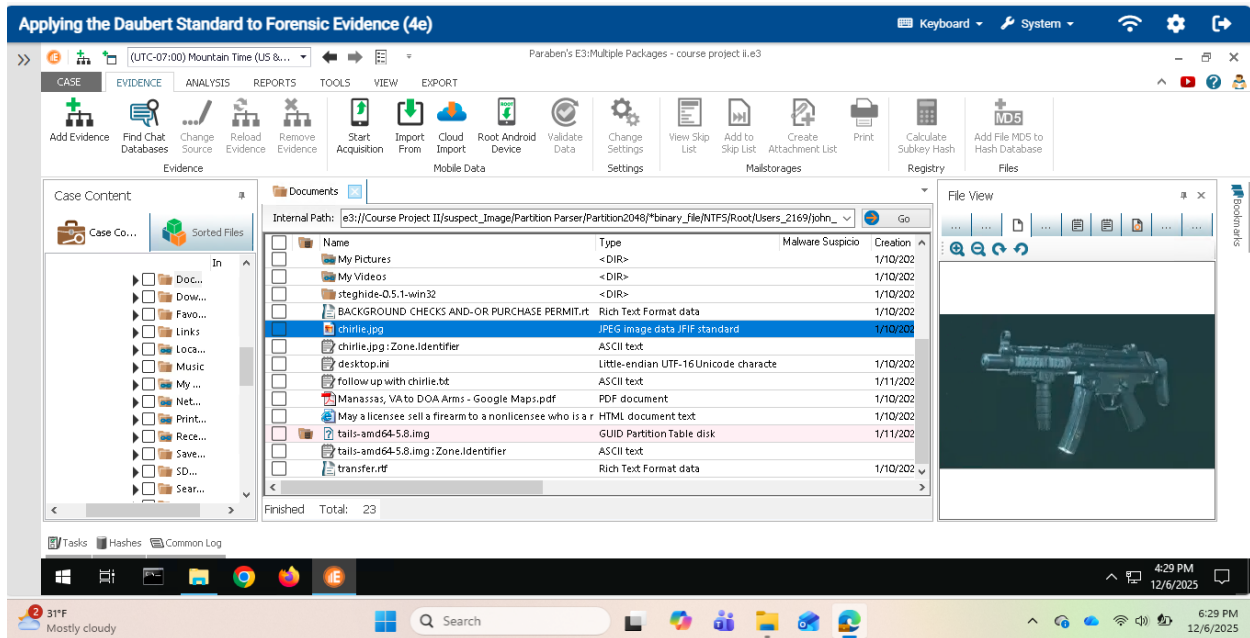
Gun Images

- A picture of multiple firearms located in **Downloads** and referenced in recently used files.

Relevance:

Gun images could be used in conjunction with Steghide to hide transaction details or personal data.





2.4 Documents Containing Transaction Discussion

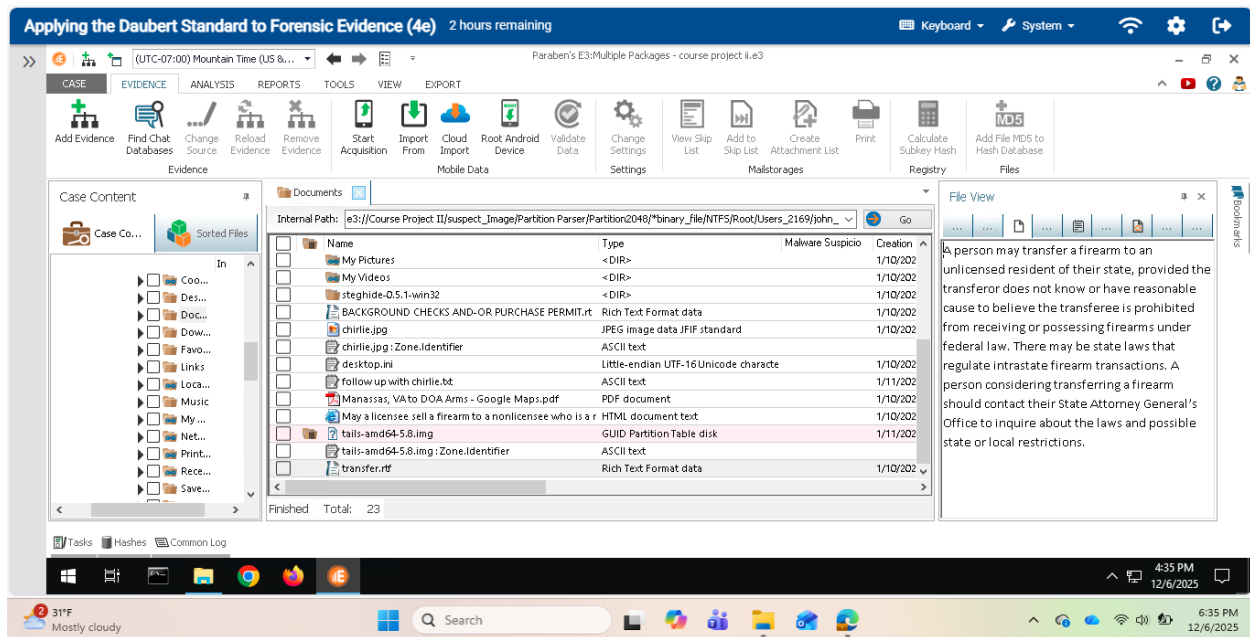
File: transfer.rtf

Inside the document:

- Discussion about **transferring a firearm to an unlicensed resident**.
- Advises contacting Attorney General office to understand legal restrictions.

Relevance:

This document directly references **interstate firearm transfer**, supporting the theory of illegal sale or avoidance of proper documentation.

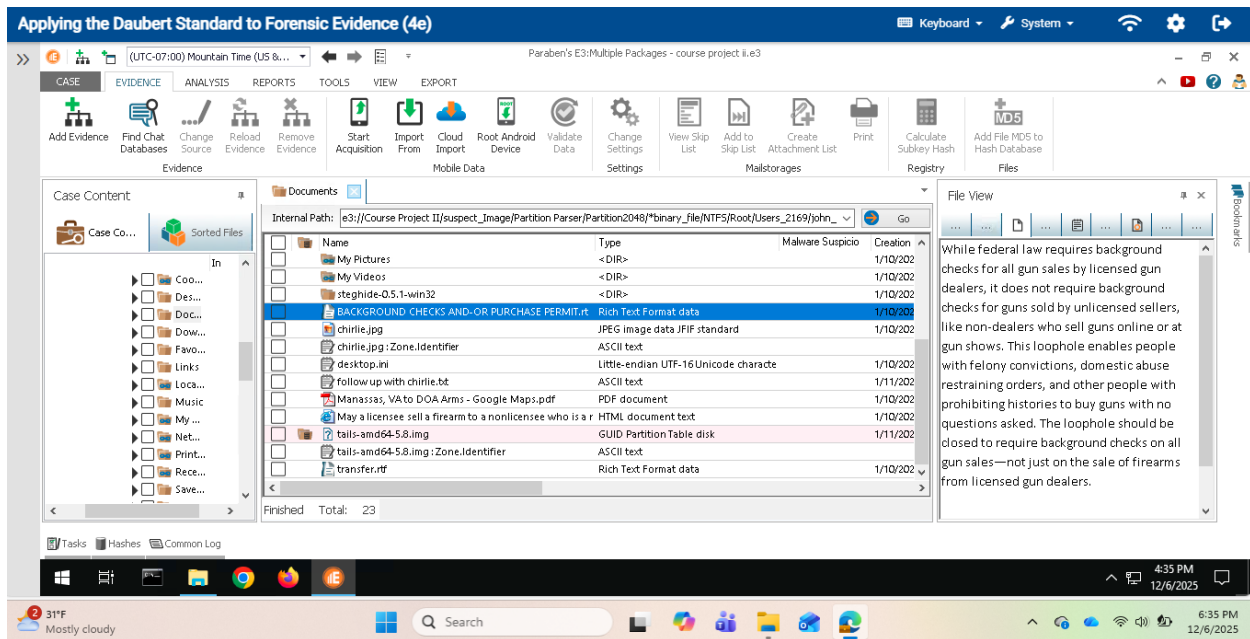


File: BACKGROUND CHECKS AND PURCHASE PERMIT.rtf

Summaries laws regarding selling firearms to nonlicensed individuals.

Relevance:

User was actively researching legal loopholes or confirming consequences regarding transferring guns to unlicensed persons.



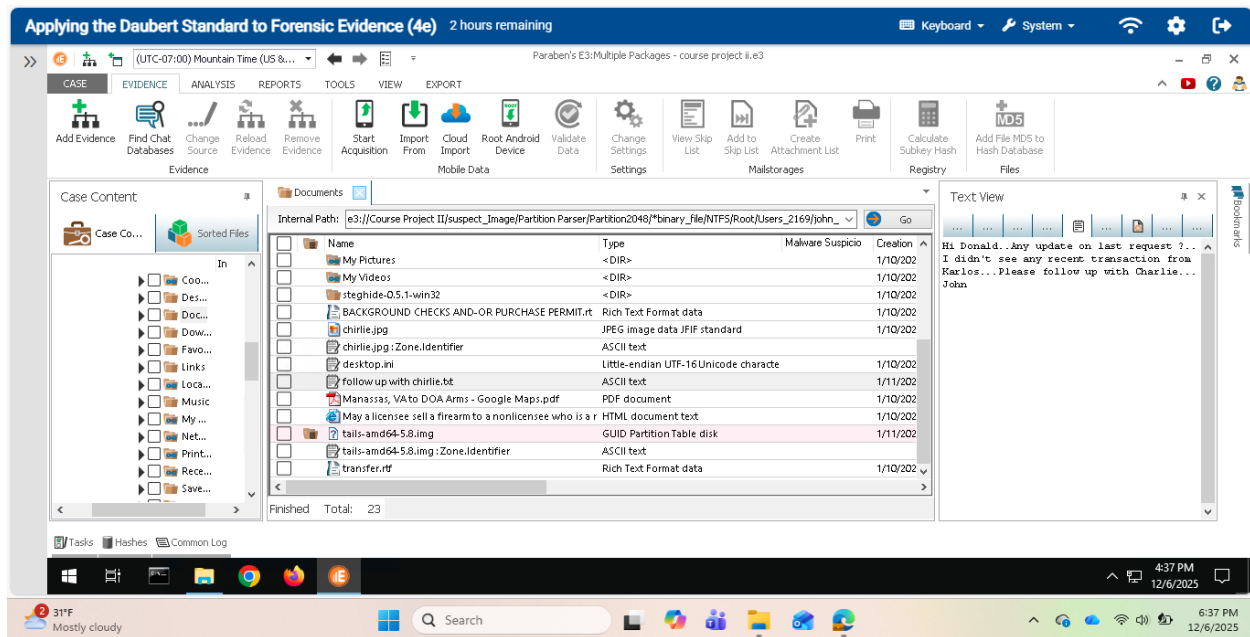
2.5 Communication Notes

Inside **follow up with Charlie.txt**:

- Mentions individuals named **Charlie, Donald, and Karlos**.
- Refers to following up regarding a transaction.

Relevance:

These individuals appear to be **associates or participants in the suspected firearm transaction**.

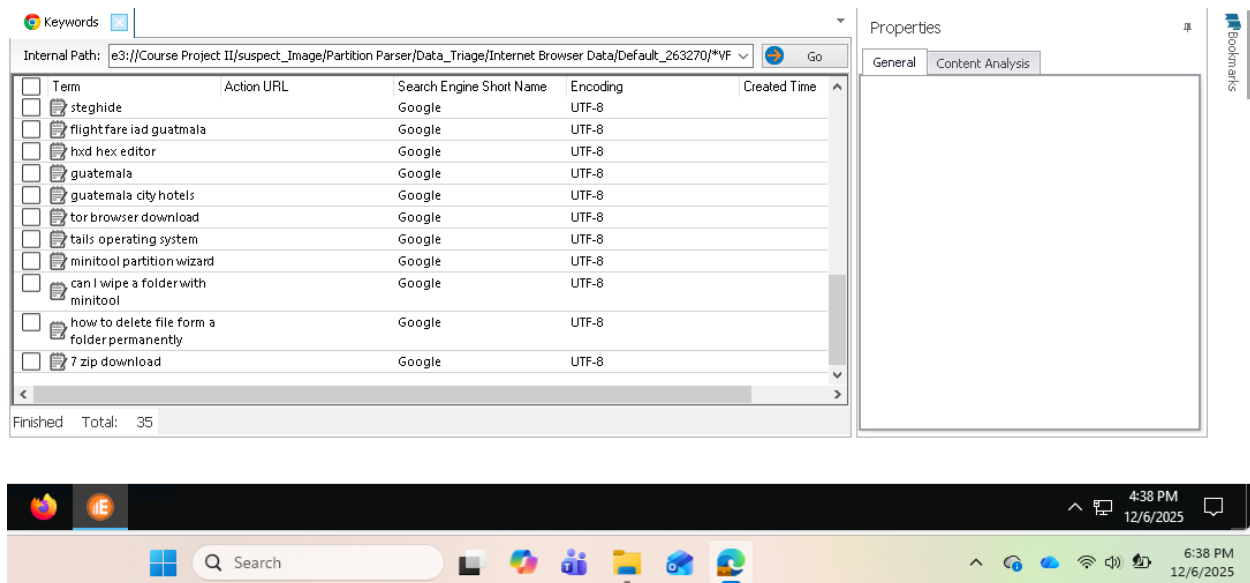


2.6 System Tools Suggesting Evidence Tampering

- **hxd hex editor** downloaded
- **minitool partition wizard** search
- Searches for deleting files permanently

Relevance:

These tools suggest the user may have attempted to **scrub** or **obfuscate** digital evidence.



3. Timeline Reconstruction

Date	Artifact	Significance
1/10/2023 12:11pm- 3:23pm	Google searches of gun shops, prices, transfer laws	Establishes planning period
1/10/2023 12:16pm	Searching for a Private Email	To communicate with the client and his partner
1/10/2023 12:16pm- 12:26pm	Searches for landscaping and security jobs around Manassas	Hints at his location being around Northern Virginia
1/10/2023 12:26pm	Searches for “gun transfer without tracing”	Shows intent to avoid proper legal process
1/10/2023 8:38pm	Searches and Enters ProtonEmail, a private email	Use of Private Communication
1/10/2023 8:52pm	Checked (Sent) Inbox	Where his emails to the client could have been sent
1/10/2023 *9:10pm *10:47pm- 10:52pm	Checks between his Email Inbox and Sent Inbox	When him and the client could’ve been communicating
1/10/2023 10:53pm	Download of the steghide tool	To conceal data in the gun images
1/10/2023 11:02pm- 11:03pm	Checks on flights from Washington to Guatemala	Hints at his Location
1/11/2023 8:37pm	Download of HxD Hex Editor	Indicates possible intent to modify or wipe data
1/11/2023 8:39pm	Searches Hotels in Guatemala	Could be his next location

1/11/2023 8:43pm	Downloads a picture	Could be used with Steghide
1/11/2023 8:44pm	Downloads the Tor Browser	To conceal his steps
1/11/2023 8:48pm	Downloads the Tails Operating System	Another private OS to conceal his steps
1/11/2023 10:40pm	Message to “Donald” found in RTF file	Likely date of firearm transaction dispute
1/11/2023 10:45pm- 10:48pm	Searches on ways to completely wipe a folder (MiniPartition Tool, WipeFile, 7zip)	To cover up his tracks and delete the evidence

Conclusion:

The user was actively researching, coordinating, and potentially performing an illegal transfer of firearms during this general timeframe, with **January 10th-11th, 2023** being the most concrete investigative date.

4. Analysis of Users & System Artifacts

User Identity & Interests

Based on system data:

- User showed interest in **security jobs, landscaping businesses**, and **travel to Guatemala**.
- This suggests multiple interests and potential employment searches.

Associates Identified

Names referenced:

- **John** – possible seller

- **Karlos** – likely buyer, possibly unlicensed
- **Donald** – receives instructions
- **Charlie** – part of follow-up communication chain

These names could be code names or real associates.

Behavioral Activity

User demonstrates:

- Knowledge of steganography
- Awareness of firearm transfer laws
- Attempts to research loopholes
- Access to images and documents relating to weapons

This establishes a behavioral pattern consistent with **concealment and illegal firearm activity**.

5. Legal and Investigative Relevance

Chain of Custody

- Image was acquired and analyzed in a controlled forensic environment.
- No alterations were made to the source.
- Findings were exported and documented for evidentiary use.

Legal Implications

Evidence suggests:

- Possible **illegal interstate firearm sale** (§922(a)(5))
- Attempt to **avoid background checks**
- Attempted **concealment via steganography and hex editing tools**

These findings support further criminal investigation but do not alone confirm guilt.

6. Report Organization & Format

This report adheres to professional forensic reporting standards:

- Sections clearly organized
- Findings logically structured
- Screenshots will be inserted with timestamps
- Language written for investigative clarity

7. Quality & Relevance of Screen Captures

Screenshots should include:

- **E3 Interface with timestamps**
- **Browser search term evidence**
- **Downloads folder with steghide**
- **RTF communication files**
- **Gun images**

Insert them where labeled.

8. Conclusion & Personal Verdict

Based on the available digital evidence:

Findings

- The suspect engaged in repeated research on **buying, selling, and transferring firearms**.
- Several searches were clearly aimed at **evading legal requirements** (e.g., selling without tracing).
- Tools for **concealment (Steghide)** and **data editing (HxD)** were downloaded.

- Documents and notes reference **associates** and **potential illegal transfers**.
- A message dated **January 10th, 2023 – January 11th, 2023** suggests a breakdown in a firearm transaction.

Verdict (within forensic limits)

The evidence strongly suggests involvement in **illegal firearm sale or transfer**, along with possible attempts at digital concealment. While this does not prove theft, it supports suspicion of **criminal firearm activity**.

As a forensic examiner, I provide only factual and artifact-based conclusions. Final determination of criminal liability rests with investigators and legal authorities.